

Solution to $\dot{\vec{x}} = A\vec{x}$ when A has repeated eigen values:-

Repeated eig values. $\rightarrow$ α_1 that repeats m times $\rightarrow$ Alg. mult of $\alpha_1 \rightarrow m$
 $\searrow$
 α_2 " " $(n-m)$ times $\rightarrow$ Alg. mult of $\alpha_2 \rightarrow n-m$
geo. mult. :- no. of LI eig vectors associated with eigen value.

for every $\alpha_i \rightarrow$ $\left. \begin{array}{l} \text{alg. mult} \\ \text{geo mult} \end{array} \right\}$

$\text{Max}(\sum \text{geo mult associated with all eig values}) \rightarrow 'n'$

$\Downarrow$

$\text{max (geo. mult)} = \text{alg. mult. of } \alpha_i$ $\left\{ \begin{array}{l} \text{geo. mult} = \text{alg mult } \forall \alpha_i \\ \text{geo. mult} \leq \text{ " " for at least one } \alpha_i \end{array} \right.$
 associated with α_i

lec-40

when $\text{geo. mult} = \text{alg. mult } \forall i \Rightarrow (\vec{v}_1, \dots, \vec{v}_n)$ that are linearly independent

$\Rightarrow$ fundamental set of sol's = $\{\vec{x}_1, \dots, \vec{x}_n\}$

when $\vec{x}_i = \vec{v}_i e^{\lambda_i t}$

when geo mult < alg mult :-

Start with α_i that repeats twice & only one linearly independent eigen vector associated with α_i .

$\downarrow$ A.M = 2 & G.M = 1.

(i=1)

$\vec{v}_1$

eig. values of $A = (\alpha_1, \alpha_1, \alpha_3, \dots, \alpha_n)$ & eig vectors = $(\underbrace{\vec{v}_1, \dots, \vec{v}_3}_{\substack{\downarrow \\ \text{G.M of } \alpha_i}}, \underbrace{\vec{v}_3, \dots, \vec{v}_n}_{\substack{\text{L.I bec of distinct} \\ \text{eig val.}}})$

(x)

$\downarrow$

fundamental set of sol's = $(\vec{v}_1 e^{\lambda_1 t}, \dots, \vec{v}_3 e^{\lambda_3 t}, \dots, \vec{v}_n e^{\lambda_n t})$.

$\vec{x}_t =$

check if $t \vec{v}_1 e^{\lambda_1 t}$ is a solution?

$$\dot{\vec{x}}_t = \lambda_1 t \vec{v}_1 e^{\lambda_1 t} + \vec{v}_1 e^{\lambda_1 t} \rightarrow (1)$$

$$A \vec{x}_t = t \underbrace{A \vec{v}_1}_{\lambda_1 \vec{v}_1} e^{\lambda_1 t} \rightarrow (2)$$

for $\vec{x}_t$ to be a sol $(1) = (2)$. Happens when $\vec{v}_1 = 0$ (which is not possible).

Try:- gen. eig vector $\rightarrow ?$

other way:- $\vec{x}_t = (t \vec{v}_1 + \vec{v}_1) e^{\lambda_1 t}$.

$$\vec{x}_t = (t\vec{u}_1 + \vec{u}_2)e^{\lambda_1 t}$$

$$\dot{\vec{x}}_t = \lambda_1 (t\vec{u}_1 + \vec{u}_2)e^{\lambda_1 t} + e^{\lambda_1 t}\vec{u}_1 = \lambda_1 t\vec{u}_1 e^{\lambda_1 t} + (\lambda_1 \vec{u}_2 + \vec{u}_1)e^{\lambda_1 t} \rightarrow (3)$$

$$\begin{aligned} A\vec{x}_t &= (tA\vec{u}_1 + A\vec{u}_2)e^{\lambda_1 t} \\ &= \lambda_1 t\vec{u}_1 e^{\lambda_1 t} + A\vec{u}_2 e^{\lambda_1 t} \rightarrow (4) \end{aligned}$$

for $\vec{x}_t$ to be a sol (3) = (4), $\Rightarrow A\vec{u}_2 = \lambda_1 \vec{u}_2 + \vec{u}_1 \Rightarrow (A - \lambda_1 I)\vec{u}_2 = \vec{u}_1 \rightarrow (5)$
 $\hookrightarrow$ gen. eig vector

If $\vec{u}_2$ satisfies (5), then $\vec{x}_t = (t\vec{u}_1 + \vec{u}_2)e^{\lambda_1 t}$ is a solution

fundamental set of sol's : $\{\vec{u}_1 e^{\lambda_1 t}, (t\vec{u}_1 + \vec{u}_2)e^{\lambda_1 t}, \vec{u}_3 e^{\lambda_3 t}, \dots, \vec{u}_n e^{\lambda_n t}\}$

$$\text{where } \vec{u}_2 : (A - \lambda_1 I)\vec{u}_2 = \vec{u}_1$$

Example:- $A = \begin{bmatrix} 1 & -3 \\ 3 & 1 \end{bmatrix}$ eig values = 4 alg mul of 2.

$$\text{eig vector } \vec{u}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\vec{u}_2 : (A - \lambda_1 I)\vec{u}_2 = \vec{u}_1 \Rightarrow \begin{bmatrix} -3 & -3 \\ 3 & 3 \end{bmatrix} \vec{u}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \Rightarrow \vec{u}_2 = \frac{1}{6} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\vec{x}_1 = \vec{u}_1 e^{\lambda_1 t}$$

$$\begin{aligned}\vec{x}_2 &= \underbrace{(t\vec{v}_1 + \vec{v}_2)}_{\substack{= \begin{bmatrix} t - y_6 \\ -t - y_6 \end{bmatrix} e^{\lambda_1 t}}} e^{\lambda_1 t} \\ &= \begin{bmatrix} t - y_6 \\ -t - y_6 \end{bmatrix} e^{\lambda_1 t}\end{aligned}$$

$$\vec{x}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{\lambda_1 t}$$

from (5): $\vec{v}_2: (A - \lambda_1 I) \vec{v}_2 = \vec{v}_1 \rightarrow$

$$\underline{A \vec{v}_i = \lambda_i \vec{v}_i}$$

mul on both sides with $(A - \lambda_1 I) \Rightarrow (A - \lambda_1 I)^2 \vec{v}_1 = \underbrace{(A - \lambda_1 I) \vec{v}_1}_{= \vec{0}}$

$$\vec{v}_1: (A - \lambda_1 I)^2 \vec{v}_2 = \vec{0} \rightarrow (??)$$

from the point of
view of Linear Tran
what does $\vec{v}_1$
mean?

General Case:- let λ_i repeat 'm' times.

Construct: $\vec{x}_i, \dots, \vec{x}_{i+m-1}$ elem in the fundamental set.

$$\vec{x}_i = \vec{v}_i e^{\lambda_i t}$$

$$\vec{x}_{i+1} = (t\vec{v}_i + \vec{v}_{i+1}) e^{\lambda_i t} \text{ where } \vec{v}_{i+1}: (A - \lambda_i I) \vec{v}_{i+1} = \vec{0}$$

$$\vec{x}_{i+2} = \left(\frac{t^2}{2} \vec{v}_i + t\vec{v}_{i+1} + \vec{v}_{i+2} \right) e^{\lambda_i t} \text{ where } \vec{v}_{i+2}: (A - \lambda_i I)^2 \vec{v}_{i+2} = \vec{0}$$

$\vdots$

$$\vec{x}_{i+m-1} = \left(\frac{t^{m-1}}{(m-1)!} \vec{v}_i + \frac{t^{m-2}}{(m-2)!} \vec{v}_{i+1} + \dots + \vec{v}_{i+m-1} \right) e^{\lambda_i t} \text{ where } \vec{v}_{i+m-1}: (A - \lambda_i I)^m \vec{v}_{i+m-1} = \vec{0}$$